

Chỉ mục

- accuracy, 415
- activation, 400
- activation function, 401
- additive, 11, 94–98, 110–111
- additivity, 305, 306
- adjusted R^2 , 87, 231, 232, 236–238
- Advertising data set, 15, 16, 19, 69, 71–73, 77, 78, 80, 82, 83, 85, 87–90, 95, 96, 109–111
- agglomerative clustering, 525
- Akaike information criterion, 87, 231, 232, 236–238
- alternative hypothesis, 76, 559
- analysis of variance, 312
- ANOVA, 587
- area under the curve, 155, 486–487
- argument, 40
- array, 42
- attribute, 42
- AUC, 155
- Auto data set, 12, 66, 98–101, 129, 197, 202–207, 327, 398
- auto-correlation, 421
- autoregression, 423
- axes, 48
- backfitting, 307, 328
- backpropagation, 429
- backward stepwise selection, 87, 234–235
- bag-of- n -grams, 415
- bag-of-words, 414
- bagging, 11, 24, 331, 343–346, 354, 360–361
- BART, 343, 350, 353, 354, 362–363
- baseline, 93, 145, 161
- basis function, 293–294, 296
- Bayes
 - classifier, 35–37, 147
 - decision boundary, 148
 - error, 35–37
- Bayes’ theorem, 146, 250
- Bayesian, 250–251, 353
- Bayesian additive regression trees,
 - 331, 343, 350, 350, 353, 354, 362–363
- Bayesian information criterion, 87, 231, 232, 236–238
- Benjamini–Hochberg procedure, 575–577
- Bernoulli distribution, 172
- best subset selection, 231, 246
- bias, 31–34, 74, 90, 159, 405
- bias-variance
 - decomposition, 32

- trade-off, 31–34, 38, 111–112, 157, 159, 163, 164, 242, 254, 263, 266, 301, 336, 376, 385
- bidirectional, 425
- Bikeshare** data set, 12, 167–172
- binary, 27, 138
- biplot, 507, 508
- Bonferroni method, 575–577, 585
- Boolean, 53, 176
- boosting, 11, 24, 331, 343, 347–350, 354, 361–362
- bootstrap, 11, 201, 212–214, 343
- Boston** data set, 12, 67, 117, 122, 133, 199, 227, 287, 327, 364, 556
- bottom-up clustering, 525
- boxplot, 62
- BrainCancer** data set, 12, 472–474, 476, 482
- branch, 333
- burn-in, 352
- C-index, 487
- Caravan** data set, 12, 184, 366
- Carseats** data set, 12, 126, 130, 364
- categorical, 2, 27
- censored data, 469–502
- censoring
 - independent, 471
 - interval, 471
 - left, 471
 - mechanism, 471
 - non-informative, 471
 - right, 471
 - time, 470
- chain rule, 429
- channel, 407
- CIFAR100** data set, 406, 409–411, 448, 449
- classification, 2, 11, 27, 34–39, 135–199, 367–382
 - error rate, 338
 - tree, 337–341, 355–358
- classifier, 135
- cluster analysis, 25–26
- clustering, 4, 25–26, 520–535
 - agglomerative, 525
 - bottom-up, 525
 - hierarchical, 521, 525–535
 - K -means, 11, 521–524
- Cochran–Mantel–Haenszel test, 475
- coefficient, 71
- College** data set, 12, 65, 286, 328
- collinearity, 106–110
- concatenation, 41
- conditional probability, 35
- confidence interval, 75–76, 90, 110, 292
- confounding, 144
- confusion matrix, 153, 176
- continuous, 2
- contour, 246
- contour plot, 50
- contrast, 94
- convenience function, 53
- convolution filter, 407
- convolution layer, 407
- convolutional neural network, 406–413
- correlation, 79, 82–83, 530
- count data, 167, 170
- Cox’s proportional hazards model, 480, 483–486
- C_p , 87, 231, 232, 236–238
- Credit** data set, 12, 91, 92, 94, 97, 98, 106–109
- cross-entropy, 405
- cross-validation, 11, 31, 34, 201–211, 231, 252, 270
 - k -fold, 206–209
 - leave-one-out, 204–206
- curse of dimensionality, 115, 193, 266
- data augmentation, 411
- data frame, 55
- Data sets
 - Advertising**, 15, 16, 19, 69, 71–73, 77, 78, 80, 82, 83, 85, 87–90, 95, 96, 109–111
 - Auto**, 12, 66, 98–101, 129, 197, 202–207, 327, 398
 - Bikeshare**, 12, 167–172

- Boston**, 12, 67, 117, 122, 133, 199, 227, 287, 327, 364, 556
- BrainCancer**, 12, 472–474, 476, 482
- Caravan**, 12, 184, 366
- Carseats**, 12, 126, 130, 364
- CIFAR100**, 406, 409–411, 448, 449
- College**, 12, 65, 286, 328
- Credit**, 12, 91, 92, 94, 97, 98, 106–109
- Default**, 12, 136–139, 141–144, 152–156, 160, 161, 225, 226, 466
- Fund**, 12, 567–570, 572, 575, 576, 585, 588, 589
- Heart**, 339, 340, 344–347, 352, 353, 382, 383
- Hitters**, 12, 332, 333, 336, 338, 339, 366, 425, 426, 437, 446
- IMDb**, 413, 415, 416, 418, 420, 437, 458, 467
- Income**, 16–18, 21–23
- Khan**, 12, 579–581, 583, 590, 593
- MNIST**, 402–404, 406, 430, 431, 441, 444, 445, 448
- NCI60**, 4, 5, 12, 546, 548–550
- NYSE**, 12, 422–424, 466, 467
- OJ**, 12, 365, 398
- Portfolio**, 12
- Publication**, 12, 482–487
- Smarket**, 2, 3, 12, 173, 184, 196
- USArrests**, 12, 507, 508, 510, 512, 513, 515, 516, 518, 519
- Wage**, 1, 2, 8, 9, 12, 290, 291, 293, 295, 297–300, 302–306, 309, 315, 327
- Weekly**, 12, 196, 226
- data type, 42
- decision function, 387
- decision tree, 11, 331–342
- deep learning, 399
- Default** data set, 12, 136–139, 141–144, 152–156, 160, 161, 225, 226, 466
- degrees of freedom, 30, 266, 295, 296, 301
- dendrogram, 521, 525–530
- density function, 146
- dependent variable, 15
- derivative, 296, 300
- detector layer, 410
- deviance, 232
- dictionary, 66
- dimension reduction, 230, 253–262
- discriminant function, 149
- discriminant method, 146–161
- dissimilarity, 530–532
- distance
 - correlation-based, 530–532, 554
 - Euclidean, 509, 522, 523, 529–532
- double descent, 431–435
- double-exponential distribution, 251
- dropout, 406, 431
- dummy variable, 91–94, 138, 142, 292
- early stopping, 430
- effective degrees of freedom, 301
- eigen decomposition, 506, 516
- elbow, 548
- embedding, 418
- embedding layer, 419
- ensemble, 343–354
- entropy, 337–339, 363
- epochs, 430
- error
 - irreducible, 17, 30
 - rate, 34
 - reducible, 17
 - term, 16
- Euclidean distance, 509, 522, 523, 529–532, 554
- event time, 470
- exception, 45
- expected value, 18
- exploratory data analysis, 504
- exponential, 173
- exponential family, 173
- F-statistic, 84

- factor, 92
- factorial, 170
- failure time, 470
- false
 - discovery proportion, 155, 573
 - discovery rate, 558, 573–577, 579–582
 - negative, 155, 562
 - positive, 155, 562, 563
 - positive rate, 155, 156, 382
- family-wise error rate, 565–573, 577
- feature, 15
- feature map, 406
- feature selection, 230
- featurize, 414
- feed-forward neural network, 400
- figure, 48
- fit, 21
- fitted value, 101
- flattening, 424
- flexible, 21
- floating point, 43
- forward stepwise selection, 86, 87, 233–234, 268
- function, 40
- Fund** data set, 12, 567–570, 572, 575, 576, 585, 588, 589
- Gamma, 173
- Gaussian (normal) distribution, 146, 147, 150, 172, 561
- generalized additive model, 5, 24, 162, 289, 290, 305–309, 319
- generalized linear model, 5, 135, 167–174, 217
- generative model, 146–161
- Gini index, 337–339, 345, 346, 363
- global minimum, 427
- gradient, 428
- gradient descent, 427
- Harrell’s concordance index, 487
- hazard function, 476–478
 - baseline, 478
- hazard rate, 476
- Heart** data set, 339, 340, 344–347, 352, 353, 382, 383
- heatmap, 50
- helper, 311
- heteroscedasticity, 103, 168
- hidden layer, 400
- hidden units, 400
- hierarchical clustering, 525–530
 - dendrogram, 525–528
 - inversion, 529
 - linkage, 529–530
- hierarchical principle, 96
- high-dimensional, 86, 234, 263
- hinge loss, 385
- Hitters** data set, 12, 332, 333, 336, 338, 339, 366, 425, 426, 437, 446
- hold-out set, 202
- Holm’s method, 568, 576, 585
- hypergeometric distribution, 501
- hyperparameter, 187
- hyperplane, 367–372
- hypothesis test, 76–77, 84, 103, 558–583
- IMDb** data set, 413, 415, 416, 418, 420, 437, 458, 467
- imputation, 515
- Income** data set, 16–18, 21–23
- increment, 60
- independent variable, 15
- indexable, 186
- indicator function, 292
- inference, 17, 18
- inner product, 379, 380
- input layer, 400
- input variable, 15
- integral, 301
- interaction, 70, 89, 95–98, 110–111, 308
- intercept, 71, 72
- interpolate, 432
- interpretability, 229
- inversion, 529
- irreducible error, 17, 36, 90, 110
- iterator, 312
- joint distribution, 158
- K-means clustering, 11, 521–524
- K-nearest neighbors, 135, 164–167
 - classifier, 11, 36–37
 - regression, 111–115

- Kaplan–Meier survival curve, 472–474, 483
- kernel, 379–382, 384, 394
 - linear, 380
 - non-linear, 377–382
 - polynomial, 380, 382
 - radial, 381–383, 390
- kernel density estimator, 159
- keyword, 46
- Khan** data set, 12, 579–581, 583, 590, 593
- knot, 290, 294, 296–299
- ℓ_1 norm, 244
- ℓ_2 norm, 242
- lag, 422
- Laplace distribution, 251
- lasso, 11, 24, 244–251, 265–266, 336, 385, 484
- leaf, 333, 526
- learning rate, 429
- least squares, 5, 21, 71–72, 140, 141, 229
 - line, 73
 - weighted, 103
- level, 92
- leverage, 104–106
- likelihood function, 141
- linear, 2, 69–115
- linear combination, 128, 230, 253, 505
- linear discriminant analysis, 5, 11, 135, 138, 147–155, 164–167, 377, 382
- linear kernel, 380
- linear model, 20, 69–115
- linear regression, 5, 11, 69–115, 172–173
 - multiple, 80–90
 - simple, 70–80
- link function, 172, 173
- linkage, 529–530, 548
 - average, 529–530
 - centroid, 529–530
 - complete, 526, 529–530
 - single, 529–530
- list, 41
- list comprehension, 123
- local minimum, 427
- local regression, 290
- log odds, 145
- log-rank test, 474–476, 483
- logistic function, 139
- logistic regression, 5, 11, 25, 135, 138–144, 164–167, 172–173, 308–309, 377, 384–385
 - multinomial, 145, 163
 - multiple, 142–144
- logit, 140
- loss function, 300, 385
- low-dimensional, 262
- LSTM RNN, 420
- main effects, 96
- majority vote, 344
- Mallow’s C_p , 87, 231, 232, 236–238
- Mantel–Haenszel test, 475
- margin, 370, 385
- marginal distribution, 158
- Markov chain Monte Carlo, 353
- matrix completion, 515
- matrix multiplication, 10
- maximal margin
 - classifier, 367–372
 - hyperplane, 370
- maximum likelihood, 139–141, 143, 170
- mean squared error, 28
- mesh, 53
- method, 43
- minibatch, 429
- misclassification error, 35
- missing at random, 515
- missing data, 56, 515–520
- mixed selection, 87
- MNIST** data set, 402–404, 406, 430, 431, 441, 444, 445, 448
- model assessment, 201
- model selection, 201
- module, 42
- multicollinearity, 108, 266
- multinomial logistic regression, 145, 163
- multiple testing, 557–583
- multi-task learning, 403
- multivariate Gaussian, 150

- multivariate normal, 150
- naive Bayes, 135, 158–161, 164–167
- namespace, 116
- natural spline, 297, 298, 301, 317
- NCI60 data set, 4, 5, 12, 546, 548–550
- negative binomial, 173
- negative predictive value, 155, 156
- neural network, 5, 399
- node
 - internal, 333
 - purity, 337–339
 - terminal, 333
- noise, 21, 252
- non-linear, 2, 11, 289–329
 - decision boundary, 377–382
 - kernel, 377–382
- non-parametric, 20, 22–23, 111–115, 193
- normal (Gaussian) distribution, 146, 147, 150, 172, 476, 561
- notebook, 40
- null, 152
 - distribution, 561, 578
 - hypothesis, 76, 559
 - model, 87, 231, 245
- null rate, 186
- NYSE data set, 12, 422–424, 466, 467
- Occam’s razor, 426
- odds, 140, 145, 195
- OJ data set, 12, 365, 398
- one-hot encoding, 92, 126, 403
- one-standard-error rule, 240
- one-versus-all, 384
- one-versus-one, 384
- one-versus-rest, 384
- optimal separating hyperplane, 370
- optimism of training error, 30
- ordered categorical variable, 315
- orthogonal, 257, 506
 - basis, 125
- out-of-bag, 345
- outlier, 103–104
- output variable, 15
- over-parametrized, 465
- overdispersion, 172
- overfitting, 21, 23, 25, 30–31, 88, 152, 233, 371
- p-value, 77, 82, 560–562, 578–579
 - adjusted, 586
- package, 42
- parameter, 71
- parametric, 20–22, 111–115
- partial least squares, 254, 260–262, 282
- partial likelihood, 480
- path algorithm, 249
- permutation, 578
- permutation approach, 577–582
- perpendicular, 257
- Poisson distribution, 169, 172
- Poisson regression, 135, 167–173
- polynomial
 - kernel, 380, 382
 - regression, 98–99, 289–292, 294–295
- pooling, 410
- population regression line, 73
- Portfolio data set, 12
- positive predictive value, 155, 156
- posterior
 - distribution, 251
 - mode, 251
 - probability, 147
- power, 108, 155, 563
- precision, 155
- prediction, 17
 - interval, 90, 110
- predictor, 15
- principal components, 505
 - analysis, 11, 254–260, 504–515
 - loading vector, 505, 506
 - missing values, 515–520
 - proportion of variance explained, 510–515, 547
 - regression, 11, 254–260, 280–282, 504, 515
 - score vector, 506
 - scree plot, 514–515
- prior
 - distribution, 251
 - probability, 146
- probability density function, 477, 478

- projection, 230
- proportional hazards assumption, 478
- pruning, 336
 - cost complexity, 336
 - weakest link, 336
- Publication data set, 12, 482–487
- Python objects and functions
 - %capture, 458
 - iloc[], 58
 - loc[], 57
 - AgglomerativeClustering(), 543
 - anova(), 313
 - anova_lm(), 125, 129, 312, 313
 - axhline(), 122, 551
 - axline(), 121, 129, 329
 - BART(), 362
 - biplot, 537
 - boot_SE(), 223
 - boxplot(), 62, 66
 - bs(), 315, 327
 - BSpline(), 315
 - clone(), 222
 - columns.drop(), 122
 - compute_linkage(), 544
 - confusion_table(), 176
 - contour(), 50
 - corr(), 129, 174
 - cost_complexity_pruning_path(), 357
 - CoxPHFitter(), 491
 - cross_val_predict(), 270
 - cross_validate(), 218, 219, 226
 - cumsum(), 539
 - cut_tree(), 545
 - data.frame(), 227
 - Dataset, 440
 - decision_function(), 392
 - DecisionTreeClassifier(), 354, 355
 - DecisionTreeRegressor(), 354
 - def, 121
 - dendrogram(), 544
 - describe(), 62, 66
 - dir(), 116
 - drop(), 179
 - dropna(), 56, 268, 461
 - DTC(), *see* DecisionTreeClassifier()
 - DTR(), *see* DecisionTreeRegressor()
 - dtype, 43
 - ElasticNetCV(), 279
 - enumerate(), 217
 - export_text(), 356
 - export_tree(), 365
 - fit(), 118, 181, 218
 - fit_transform(), 119
 - for, 59
 - GaussianNB(), 182
 - GBR(), *see* GradientBoostingRegressor()
 - get_dummies(), 461
 - get_influence(), 121
 - get_prediction(), 120, 314
 - get_rdataset(), 535
 - glm(), 313
 - glob(), 437
 - GradientBoostingClassifier(), 361
 - GradientBoostingRegressor(), 354, 361
 - GridSearchCV(), 276
 - groupby(), 490
 - hist(), 62
 - iloc[], 58, 59
 - import, 42
 - imshow(), 50, 449
 - ISLP.bart, 362
 - ISLP.cluster, 544
 - json, 437
 - KaplanMeierFitter(), 502
 - keras, 437
 - KFold(), 219
 - KMeans(), 542, 543
 - Kmeans(), 542
 - KNeighborsClassifier(), 183
 - lambda, 58
 - LDA(), *see* LinearDiscriminantAnalysis()
 - legend(), 132
 - lifelines, 490
 - LinearDiscriminantAnalysis(), 174, 179
 - LinearGAM(), 317
 - LinearRegression(), 280

- `load_data()`, 117
- `loc[]`, 58, 59, 177
- `log_loss()`, 355
- `LogisticGAM()`, 323
- `logrank_test()`, 490
- `lowess()`, 324
- `matplotlib`, 48
- `max()`, 66
- `mean()`, 48
- `median()`, 197
- `min()`, 66
- `MNIST()`, 444
- `ModelSpec()`, 116–118, 122, 124, 267
- `MS()`, *see* `ModelSpec()`
- `mult_test()`, *see* `multipletests()`
- `multipletests()`, 586
- `multipletests()`, 583, 589
- `multivariate_logrank_test()`, 496
- `NaturalSpline()`, 317, 319
- `ndim`, 42
- `nn.RNN()`, 461
- `normal()`, 132, 286, 555
- `np`, *see* `numpy`
- `np.all()`, 54, 180
- `np.allclose()`, 190
- `np.any()`, 54
- `np.arange()`, 51
- `np.argmax()`, 122
- `np.array()`, 42
- `np.concatenate()`, 133
- `np.corrcoef()`, 46, 554
- `np.empty()`, 224
- `np.isnan()`, 268
- `np.ix_()`, 53
- `np.linalg.svd()`, 539
- `np.linspace()`, 50
- `np.logspace()`, 318
- `np.mean()`, 47, 176
- `np.nan`, 60
- `np.nanmean()`, 541
- `np.percentile()`, 228
- `np.power()`, 219
- `np.random.choice()`, 553
- `np.random.default_rng()`, 46, 47
- `np.random.normal()`, 45
- `np.sqrt()`, 45
- `np.squeeze()`, 457
- `np.std()`, 47
- `np.sum()`, 43
- `np.var()`, 47
- `np.where()`, 180
- `ns()`, 317
- `numpy`, 42, 555
- `os.chdir()`, 55
- `outer()`, 219
- `pairwise_distances()`, 554
- `pairwise_tukeyhsd()`, 587
- `pandas`, 55
- `params`, 175
- `partial()`, 222, 269
- `PCA()`, 280, 537, 540, 554
- `pd`, *see* `pandas`
- `pd.crosstab()`, 555
- `pd.cut()`, 315
- `pd.get_dummies()`, 314
- `pd.plotting.scatter_matrix()`, 62
- `pd.qcut()`, 314, 315
- `pd.read_csv()`, 55, 556
- `pd.Series()`, 62
- `Pipeline()`, 275
- `plot()`, 48, 61, 356, 490
- `plot.scatter()`, 120
- `plot_gam()`, 321
- `plot_svm()`, 398
- `PLSRegression()`, 282
- `poly()`, 125, 313, 327
- `predict()`, 175, 178, 181, 216, 218, 323, 358
- `predict_survival_function()`, 493
- `print()`, 40
- `pvalues`, 175
- `pygam`, 307, 317
- `pytorch_lightning`, 435
- `QDA()`, *see* `QuadraticDiscriminantAnalysis()`
- `QuadraticDiscriminantAnalysis()`, 174, 181
- `random()`, 555
- `RandomForestRegressor()`, 354, 360
- `read_image()`, 436
- `reindex()`, 461
- `reshape()`, 43

- `return`, 198
- `RF()`, *see* `RandomForestRegressor()`
- `rng`, *see* `np.random.default_rng()`
- `rng.choice()`, 60
- `rng.standard_normal()`, 60
- `roc_curve()`, 392
- `RocCurveDisplay.from_estimator()`, 387
- `savefig()`, 50
- `scatter()`, 49, 61
- `scipy.interpolate`, 315
- `score()`, 218, 461
- `seed_everything()`, 436
- `set_index()`, 57
- `set_title()`, 49
- `set_xlabel()`, 49
- `set_xscale()`, 198
- `set_ylabel()`, 49
- `set_yscale()`, 198
- `shape`, 43
- `ShuffleSplit()`, 219
- `sim_time()`, 495
- `SimpleDataModule()`, 441
- `SimpleModule.classification()`, 446
- `SimpleModule.regression()`, 442
- `skl`, *see* `sklearn.linear_model`
- `skl.ElasticNet()`, 273, 277
- `skl.ElasticNet.path`, 274
- `skl.ElasticNet.path()`, 273
- `sklearn`, 118, 181
- `sklearn.ensemble`, 360
- `sklearn.linear_model`, 267
- `sklearn.model_selection`, 267
- `sklearn_selected()`, 269
- `sklearn_selection_path()`, 270
- `sklearn_sm()`, 218
- `skm`, *see* `sklearn.model_selection`
- `skm.cross_val_predict()`, 271
- `skm.KFold()`, 271
- `skm.ShuffleSplit()`, 272
- `slice()`, 51, 462
- `sm`, *see* `statsmodels`
- `sm.GLM()`, 174, 192, 226
- `sm.Logit()`, 174
- `sm.OLS()`, 118, 129, 174, 319
- `StandardScaler()`, 185, 438, 537, 555
- `statsmodels`, 116, 173
- `std()`, 186
- `Stepwise()`, 269
- `str.contains()`, 59
- `subplots()`, 48
- `sum()`, 43, 268
- `summarize()`, 118, 129, 223, 226
- `summary()`, 119, 322, 587
- `super()`, 440
- `SupportVectorClassifier()`, 387, 389–391, 393
- `SupportVectorRegression()`, 394
- `SVC()`, *see* `SupportVectorClassifier()`
- `svd()`, 539
- `SVR()`, *see* `SupportVectorRegression()`
- `TensorDataset()`, 441
- `to_numpy()`, 437
- `torch`, 435
- `torchinfo`, 436
- `torchmetrics`, 436
- `torchvision`, 436
- `ToTensor()`, 444
- `train_test_split()`, 186, 216
- `transform()`, 118, 119
- `ttest_1samp()`, 584
- `ttest_ind()`, 590
- `ttest_rel()`, 587
- `tuple`, 43
- `uniform()`, 555
- `value_counts()`, 66
- `var()`, 536
- `variance_inflation_factor()`, 116, 124
- `VIF()`, *see* `variance_inflation_factor()`
- `where()`, 355
- `zip()`, 60, 312
- q-values, 589
- quadratic, 98
- quadratic discriminant analysis, 4, 135, 156–157, 164–167

- qualitative, 2, 27, 91, 135, 167, 202
 - variable, 91–94
- quantitative, 2, 27, 91, 135, 167, 202
- radial kernel, 381, 383, 390
- random forest, 11, 331, 343, 346–347, 354, 360–361
- random seed, 46
- re-sampling, 577–582
- recall, 155
- receiver operating characteristic (ROC), 154, 382–383
- recommender systems, 516
- rectified linear unit, 401
- recurrent neural network, 416–427
- recursive binary splitting, 334, 337, 338
- reducible error, 17, 90
- regression, 2, 11, 27
 - local, 289, 290, 304–305
 - piecewise polynomial, 294–295
 - polynomial, 289–292, 299
 - spline, 289, 294
 - tree, 331–337, 358–360
- regularization, 230, 240, 406, 484–486
- ReLU, 401
- resampling, 201–214
- residual, 71, 81
 - plot, 100
 - standard error, 75, 77–78, 88–89, 109
 - studentized, 104
 - sum of squares, 71, 79, 81
- residuals, 263, 348
- response, 15
- ridge regression, 11, 240–244, 385, 484
- risk set, 473
- robust, 374, 376, 535
- ROC curve, 154, 382–383, 486–487
- R^2 , 77–80, 88, 109, 238
- rug plot, 314
- scale equivariant, 242
- Scheffé’s method, 572
- scree plot, 512, 514–515
- elbow, 514
- semi-supervised learning, 27
- sensitivity, 153, 155, 156
- separating hyperplane, 367–372
- Seq2Seq, 425
- sequence, 41
- shrinkage, 230, 240, 484–486
 - penalty, 240
- sigmoid, 401
- signal, 252
- signature, 45
- singular value decomposition, 539
- slack variable, 375
- slice, 51
- slope, 71, 72
- Smarket** data set, 2, 3, 12, 173, 184, 196
- smoother, 308
- smoothing spline, 290, 300–303
- soft margin classifier, 372–374
- soft-thresholding, 250
- softmax, 145, 405
- sparse, 244, 252
- sparse matrix format, 414
- sparsity, 244
- specificity, 153, 155, 156
- spline, 289, 294–303
 - cubic, 296
 - linear, 296
 - natural, 297, 301
 - regression, 289, 294–299
 - smoothing, 30, 290, 300–303
 - thin-plate, 22
- standard error, 75, 101
- standardize, 185
- statistical model, 1
- step function, 111, 289, 292–293
- stepwise model selection, 11, 231, 233
- stochastic gradient descent, 429
- string, 41
- string interpolation, 490
- stump, 349
- subset selection, 230–240
- subtree, 336
- supervised learning, 25–27, 261
- support vector, 371, 376, 385
 - classifier, 367, 372–377
 - machine, 5, 11, 24, 377–386

- regression, 386
- survival
 - analysis, 469–502
 - curve, 472, 483
 - function, 472
 - time, 470
- synergy, 70, 89, 95–98, 110–111
- systematic, 16
- t-distribution, 77, 165
- t-statistic, 76
- t-test
 - one-sample, 583, 584, 588
 - paired, 587
 - two-sample, 559, 570, 571, 577–581, 584, 590
- test
 - error, 35, 37, 176
 - MSE, 28–32
 - observations, 28
 - set, 30
 - statistic, 559
- theoretical null distribution, 577
- time series, 101
- total sum of squares, 79
- tracking, 102
- train, 21
- training
 - data, 20
 - error, 35, 37, 176
 - MSE, 28–31
- transformer, 311
- tree, 331–342
- tree-based method, 331
- true negative, 155
- true positive, 155
- true positive rate, 155, 156, 382
- truncated power basis, 296
- Tukey’s method, 571, 585, 587
- tuning parameter, 187, 240, 484
- two-sample *t*-test, 474
- Type I error, 155, 562–565
- Type I error rate, 563
- Type II error, 155, 563, 568, 584
- unsupervised learning, 25–27, 255, 260, 503–552
- USArrests data set, 12, 507, 508, 510, 512, 513, 515, 516, 518, 519
- validation set, 202
 - approach, 202–204
- variable, 15
 - dependent, 15
 - dummy, 91–94, 97–98
 - importance, 346, 360
 - independent, 15
 - indicator, 35
 - input, 15
 - output, 15
 - qualitative, 91–94, 97–98
 - selection, 86, 230, 244
- variance, 18, 31–34, 159
 - inflation factor, 108–110, 123
- varying coefficient model, 305
- Wage data set, 1, 2, 8, 9, 12, 290, 291, 293, 295, 297–300, 302–306, 309, 315, 327
- weak learner, 343
- weakest link pruning, 336
- Weekly data set, 12, 196, 226
- weight freezing, 412, 419
- weight sharing, 418
- weighted least squares, 103, 304
- weights, 404
- with replacement, 214
- within class covariance, 150
- wrapper, 217